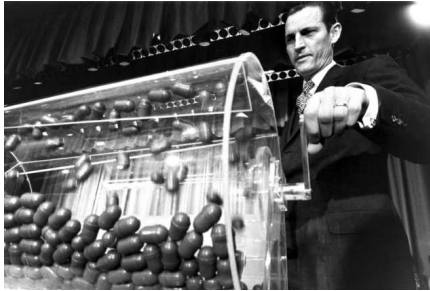
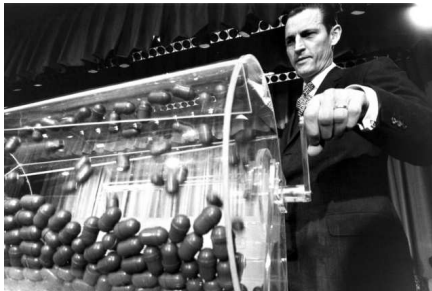


The Vietnam Draft



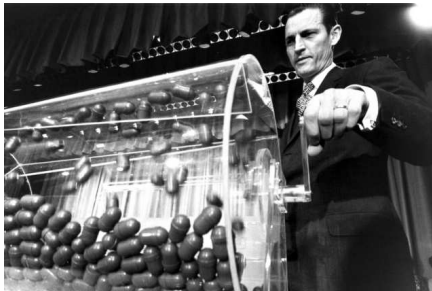
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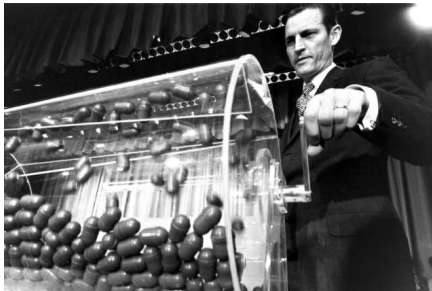
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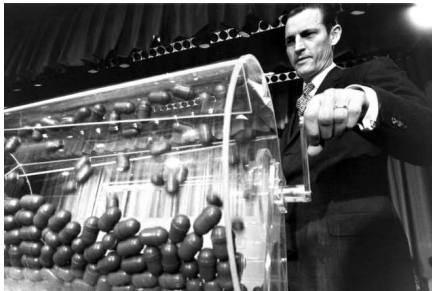
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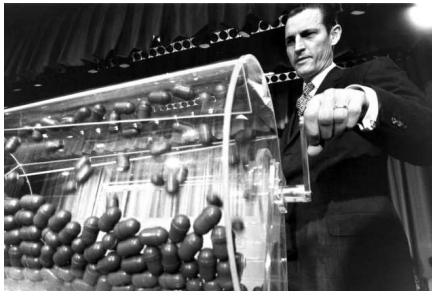
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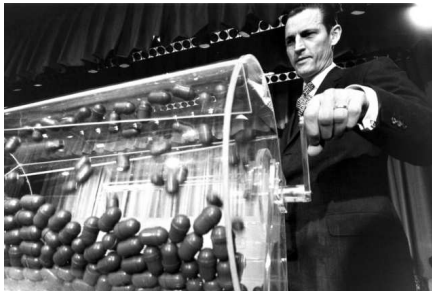
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Randomness vs. Uniformity vs. Fairness...



Cracking Scratch Off Games



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- Randomness vs. Predictability



What is Random?

00000000000000000000... ?



What is Random?

00000000000000000000101101001111... ?



What is Random?

00000000000000000000101101001111... ?

...it makes more sense,



What is Random?

00000000000000000000101101001111... ?

...it makes more sense, *theoretically*,



What is Random?

00000000000000000000101101001111... ?

...it makes more sense, *theoretically*, to restrict to *infinite* bit strings...



What is Random?

1010101010101010101010101... ?



What is Random?

1010101010101010101010101... ?

...limiting frequencies having well-defined limits is
a nice property,



What is Random?

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...limiting frequencies having well-defined limits is
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What is Random?

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...limiting frequencies having well-defined limits is a nice property, but not enough! More stringently: a random sequence should pass *statistical tests* and/or evade “simple” description.



Algorithmic Information Theory...

...has greatly clarified what randomness is.



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...has greatly clarified what randomness is. But truly understanding randomness,



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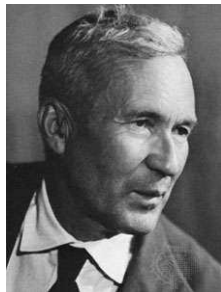
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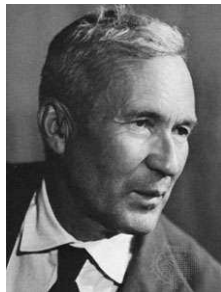
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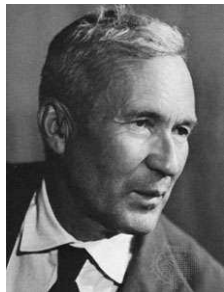
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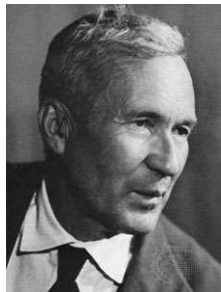
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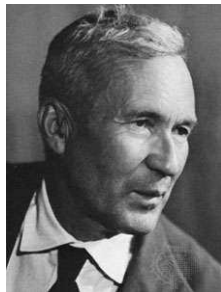
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Motivating Pseudo-Randomness #1



Suppose you ask Detective K to bet on a coin toss...



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Is the result **random**?



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Suppose you ask famed gambler and computer scientist, Chris Ferguson, to bet on a coin toss...



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Suppose you ask famed gambler and computer scientist, Chris Ferguson, to bet on a coin toss...

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- You are in a room full of motion capture cameras and displays, interfaced to a super-computer that Chris can use.



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